

Local Normalizing Flows

Tony Deng

1 Introduction

In [1], Tabak and Turner proposed to write the transportation map as a composition of elementary maps, each is a perturbation of the identity. According to this map, the position of particle i at time step $n + 1$ can be deduced from the following formula:

$$x_i^{n+1} = x_i^n + \beta f(\|x_i^n - \tilde{x}^{n+1}\|)(x_i^n - \tilde{x}^{n+1})$$

where x_i^n denote the particle's position at time step n . However, since we can not guarantee the composition of the gradient of a convex map is still the gradient of a convex map, we propose to write the transportation map as a linear combination of the elementary maps. According to this map, the particle at time step $n + 1$ can be deduced from the following formula:

$$x_i^{n+1} = x_i^n + \beta f(\|x_i^0 - \tilde{x}^{n+1}\|)(x_i^0 - \tilde{x}^{n+1})$$

We can also expand the formula and write the whole expression as

$$x_i^{n+1} = x_i^0 + \sum_{k=1}^{n+1} \beta_k f(\|x_i^0 - \tilde{x}^k\|)(x_i^0 - \tilde{x}^k)$$

Let J_n denote the Jacobian matrix of the map T at time step n . If $\tilde{x}^{n+1} = x_i^0$, then $J_{n+1} = J_n$. If otherwise, we have the following recursive formula:

$$J_{n+1} = J_n + \beta f(r_{n+1})I + \beta \frac{f'(r_{n+1})}{r_{n+1}}(x_i^0 - \tilde{x}^{n+1})(x_i^0 - \tilde{x}^{n+1})^T$$

I failed to express the determinant of J_{n+1} from the determinant of J_n by a simple, analytic formula, but the determinant of any symmetric matrix can be computed in $O(d^3)$ time. Therefore, we can store J_n as we iterate through n and compute the determinant by python.

2 Log determinant and log likelihood

I find some methods to calculate

$$\frac{\partial}{\partial \beta} \log(\det J_{n+1}) \big|_{\beta=0}$$

We can write the expression as

$$\begin{aligned} \frac{\partial}{\partial \beta} \log(\det J_{n+1}) &= \frac{\partial / \partial \beta (\det J_{n+1}) \big|_{\beta=0}}{\det J_{n+1} \big|_{\beta=0}} \\ &= \frac{\text{tr}(\text{adj}(J_n)(\partial / \partial \beta) J_{n+1} \big|_{\beta=0})}{\det J_n} \quad \text{by Jacobi's formula} \end{aligned}$$

Here, $\text{adj}(J_n)$ is the adjugate matrix of J_n . If J_n is invertible, $\text{adj}(J_n) = J_n^{-1} \det J_n$ and the expression can be written as

$$\frac{\partial}{\partial \beta} \log(\det J_{n+1}) = \text{tr} \left(J_n^{-1} \frac{\partial}{\partial \beta} J_{n+1} \big|_{\beta=0} \right)$$

$\partial / \partial \beta J_{n+1}$ is the entry-wise differentiation with respect to β . In particular, if $x_i^0 = \tilde{x}^{n+1}$, we know $\partial / \partial \beta J_{n+1} \big|_{\beta=0} = 0$ is the all zero matrix. If otherwise,

$$\frac{\partial}{\partial \beta} J_{n+1} \big|_{\beta=0} = f(r_{n+1})I + \frac{f'(r_{n+1})}{r_{n+1}}(x_i^0 - \tilde{x}^{n+1})(x_i^0 - \tilde{x}^{n+1})^T$$

In this case, therefore,

$$\frac{\partial}{\partial \beta} \log(\det J_{n+1}) \big|_{\beta=0} = \text{tr} \left(J_n^{-1} \left(fI + \frac{f'}{r}(x_i^0 - \tilde{x}^{n+1})(x_i^0 - \tilde{x}^{n+1})^T \right) \right)$$

Let μ be the pdf of independent normal distribution. We know

$$\begin{aligned} \frac{\partial}{\partial \beta} \log \mu(T(x)) \big|_{\beta=0} &= \frac{\partial}{\partial \beta} \left(-\frac{n}{2} \log(2\pi) - \frac{1}{2} \|T(x)\|^2 \right) \big|_{\beta=0} \\ &= -T(x) \big|_{\beta=0} \cdot \frac{\partial}{\partial \beta} T(x) \big|_{\beta=0} \\ &= -x_i^n \cdot \frac{\partial}{\partial \beta} x_i^{n+1} \big|_{\beta=0} \\ &= -x_i^n \cdot f(\|x_i^0 - \tilde{x}^{n+1}\|)(x_i^0 - \tilde{x}^{n+1}) \end{aligned}$$

Therefore, if $x_i^0 \neq \tilde{x}^{n+1}$, the gradient of the log likelihood is determined by

$$\begin{aligned} \frac{\partial}{\partial \beta} L &= \frac{\partial}{\partial \beta} \log(J_{n+1}) + \log \mu(T(x)) \big|_{\beta=0} \\ &= \text{tr} \left(J_n^{-1} \left(fI + \frac{f'}{r}(x_i^0 - \tilde{x}^{n+1})(x_i^0 - \tilde{x}^{n+1})^T \right) \right) - x_i^0 \cdot f(\|x_i^n - \tilde{x}^{n+1}\|)(x_i^0 - \tilde{x}^{n+1}) \end{aligned}$$

If $x_i^0 = \tilde{x}^{n+1}$, the gradient of the log likelihood is simply 0.

References

- [1] A family of nonparametric density estimation algorithms by E.G. Tabak, C.V. Turner (2013): available at <https://core.ac.uk/download/pdf/80364696.pdf>